Python for Data Science - 2305CS303

Lab - 2

Roll No. : 122

Name : Yatharth Sanghavi

1. WAP to check whether the given number is Positive or Negative.

In [2]:
```
a=int(input("enter a: "))
if a<0:
    print("number is negative")
else:
    print("number is positive")
```

```
number is positive
```

2. WAP to check whether the given number is Odd or Even.

In [3]:
```
a=int(input("enter a: "))
if a%2==0:
    print("number is even")
else:
    print("number is odd")
```

```
number is even
```

3. WAP to find out Largest number from given two numbers using simple if and ternary operator.

In [10]:
```
a=int(input("enter a: "))
b=int(input("enter b: "))
if a>b:
    print("A is greater")
else:
    print("B is greater")
c= a if a>b else b
print(c)
```

```
B is greater
20
```

4. WAP to find out Largest number from given three numbers.

In [14]:
```
a=int(input("enter a: "))
b=int(input("enter b: "))
c=int(input("enter c: "))
if a>b and a>c:
    print("a is greater")
elif b>a and b>c:
    print("b is greater")
else:
    print("c is greater")
```

5. WAP to check whether the given year is Leap year or not.

[If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year].

In [18]:
```
y=int(input("enter year"))
if (y%4==0 and y%100!=0) or y%400==0:
    print("leap year")
else:
    print("not lear year")
```

6. WAP to display the name of the Day according to the number given by the user.

In [22]:
```
day_number = int(input("Enter a day number (1-7): "))

days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sun

if 1 <= day_number <= 7:
    print("The day is:", days[day_number - 1])
else:
    print("Invalid input! Please enter a number between 1 and 7.")
```

```
The day is: Monday
```

7. WAP to implement simple Calculator which performs (add,sub,mul,div) of two numbers based on user input.

In [21]:
```
a=int(input("enter a: "))
b=int(input("enter b: "))
c=int(input("enter 1 for + , 2 for -, 3 for /, 4 for *:"))
if c==1:
    print(a+b)
elif c==2:
    print(a-b)
elif c==3:
    print(a/b)
elif c==4:
    print(a*b)
```

```
else:
    print("enter between 1 to 4")
```

```
70
```

8. WAP to calculate electricity bill based on following criteria. Which takes the unit from the user.

-First 1 to 50 units – Rs. 2.60/unit

-Next 50 to 100 units – Rs. 3.25/unit

-Next 100 to 200 units – Rs. 5.26/unit

-200 units – Rs. 8.45/unit

In [20]:
```
a=int(input("enter bill: "))
d=0
if a<=50:
    print(a*2.60)
elif a>50 and a<=100:
    print((50*2.60)+((a-50)*3.25))
elif a>100 and a<=200:
    print((50*2.60)+(50*3.25)+((a-100)*5.26))
else:
    print((50*2.60)+(50*3.25)+(50*5.26)+((a-150)*8.45))
```

```
424.0
```

9. WAP to simulate movie ticket booking system this program should accept the customer age,day of booking(weekday,weekend) and seat type(silver,gold, platinum) as input.

ticket price should be calculated according to following rule:-

1. seat type(base price:- silver seat:- 150, gold:- 250,platinum:-400)

2. if age below 12 50% discount on base price or age between 12 to 59 no discount or age above 60 discount 30% on base price

3. booking on weekend add additional 20% charge is applied to ticket price after age discount.else no additional charge applied on weekdays

In [10]:
```
age = int(input("enter age: "))
day = input("enter day of booking: ")
seattype= input("enter seat type(silver,gold, platinum): ")
price = 150 if seattype == "silver" else 250 if seattype == "gold" else 400 if s
p=price
if int(price):
    if age<12:
        dis=int(price*50/100)
        price-=dis
    elif age>=60:
        dis=int(price*30/100)
        price-=dis
    if day=="saturday" or day=="sunday":
```

```
        inc=int(price*20/100)
        price+=inc
else:
    print("enter proper seat type without typo error")
print(f"total price : {price} seat type: {seattype} base price: {p} discount: {d
```

```
total price : 180 seat type: silver base price: 150 discount: 75
```